

# SMC Virtual Platform — Exit Criteria

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# SMC Virtual Platform — Exit Criteria

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**Document scope.** This document defines the binary, auditable conditions under which the SMC SystemC/TLM-2.0 Virtual Platform (VP) is declared *complete* and released to firmware, post-silicon, and software-validation consumers as `smc-vp-1.0.0`. Each criterion is paired with a hard pass condition and the artefact that proves it.

## Companion documents.

- `01_SMC_Architecture.md` — architectural reference, IP list, modeling priority (P1/P2/P3).
- `02_SMC_IP_LowLevel_Design.md` — per-IP design, ISS integration, milestones M1–M4 (Appendix A §A.14).
- `03_SMC_Test_Plan.md` — test cases, coverage targets, regression tiers.

**Release rule.** The VP is released only when **all five gates** below are simultaneously green for two consecutive weekly regressions. Any gate failing demotes the candidate to the previous tag.

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## Gate Summary

| Gate | Theme                                          | # of criteria | Owner             |
|------|------------------------------------------------|---------------|-------------------|
| 1    | Functional completeness (model surface)        | 5             | Modeling Lead     |
| 2    | Verification quality                           | 5             | Verification Lead |
| 3    | Performance & usability                        | 5             | Modeling Lead     |
| 4    | Deliverables & process                         | 6             | Project Lead      |
| 5    | Negative criteria (explicitly accepted limits) | 6             | Architecture Lead |

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## Gate 1 — Functional Completeness

| ID | Criterion                   | Pass Condition                                                                                                                                                                                                                                | Evidence                                  |
|----|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| F1 | All P1 IPs implemented      | All 16 P1 IPs (Fabric, AXI Filter, CPU Cluster, PLIC, CLINT, Mailbox, OCTS, eFuse, UART, I2C, AVSBus, Reset Unit, Boot ROM, Scratchpad SRAM, Log Engine, I3C[0]) instantiate inside <code>smc_top</code> without <code>SC_REPORT_ERROR</code> | <code>make smc_top</code> clean build log |
| F2 | All P2 IPs implemented      | DMA, Memory Zeroer, GPIO, Telemetry, PLL, MISC, Debug Module, WDT, BEU, PVT, I3C[1..5] instantiate and pass unit tests                                                                                                                        | Tier-1 unit-test report                   |
| F3 | Memory map matches HW spec  | Every range in <code>01_SMC_Architecture.md</code> §2.2 routed by Fabric; no aliases, no gaps                                                                                                                                                 | <code>TC-FAB-MMAP-NNN</code> green        |
| F4 | Register surface complete   | Every register in test plan §B.1–B.25 responds with correct reset value and supports R/W as specified                                                                                                                                         | Coverage report: 100 % register coverage  |
| F5 | Cluster bring-up M4 reached | LLD §A.14 milestones M1 → M2 → M3 → M4 all green                                                                                                                                                                                              | M1–M4 milestone sign-off reports          |

## Gate 2 — Verification Quality

| ID | Criterion                           | Pass Condition                                                                                                                                            | Evidence                                              |
|----|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| V1 | Unit-test pass rate                 | 100 % of <code>TC-&lt;IP&gt;-NNN</code> tests in test plan §B pass                                                                                        | Tier-1 nightly green for 7 consecutive days           |
| V2 | SoC integration tests pass          | 100 % of <code>TC-S0C-*</code> tests in test plan §C pass (Boot, FLR, IRQ, DMA, OCTS, Security, Logging, Telemetry, Stress)                               | Tier-2 weekly green for 2 consecutive weeks           |
| V3 | Coverage thresholds met             | Line $\geq 95$ %; Branch $\geq 90$ %; Register = 100 %; Reset values = 100 %; IRQ-source = 100 %; Functional-coverage scenarios = 100 %                   | <code>coverage/index.html</code> — all bars green     |
| V4 | Long-soak clean                     | Tier-3 (8 h random + error injection + OCTS sync drift) with zero unexplained failures                                                                    | Tier-3 log + triage notes                             |
| V5 | Commit-log diff vs Rocket-Verilator | Lockstep run of three sign-off firmware images (boot, FLR, eFuse-read) shows zero ISA-level divergence (allow-listed Rocket-vs-Spike CSR deltas excluded) | <code>tools/diff_commits.py</code> report (LLD §A.12) |

## Gate 3 — Performance & Usability

| ID | Criterion          | Pass Condition                                                                                                           | Evidence                                 |
|----|--------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| P1 | Boot wall-clock    | Boot to <code>OCCP_READY</code> $\leq 30$ s on a single x86_64 host core in <code>SMC_TD_MODE=fast</code>                | <code>bench/boot_time.csv</code>         |
| P2 | Throughput         | $\geq 50$ MIPS aggregate across 4 Rocket harts at quantum = 1000 instructions                                            | <code>bench/mips.csv</code>              |
| P3 | IRQ-latency budget | Mean PLIC source $\rightarrow$ trap-entry $\leq 2$ $\mu$ s in fast mode; $\leq 50$ ns in lockstep mode                   | <code>TC-IRQ-LAT-NNN</code> report       |
| P4 | Determinism        | Same firmware + seed produces byte-identical commit log on Linux x86_64 and macOS arm64                                  | <code>tools/replay_diff.py</code> report |
| P5 | Build hygiene      | Zero <code>-Wall -Wextra -Wpedantic</code> warnings; zero new TSan/UBSan findings; zero new clang-tidy critical findings | CI build matrix                          |

## Gate 4 — Deliverables & Process

| ID | Criterion              | Pass Condition                                                                                                                                                                         | Evidence                                            |
|----|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| D1 | Documentation in sync  | Architecture, LLD and Test Plan PDFs regenerated; every IP has a §5.x description and a §B.x test plan                                                                                 | PDFs dated within $\leq 7$ days of candidate tag    |
| D2 | API stability declared | <code>iss_hart</code> , <code>smc_axi_extension</code> , public sockets of <code>smc_top</code> carry <code>SMC_VP_API_VERSION = "1.0"</code> and a documented backwards-compat policy | <code>include/smc/version.h</code> + CHANGELOG      |
| D3 | Spike submodule pinned | <code>external/spike</code> SHA recorded in lockfile; local patches in <code>external/spike-cmake/patches/</code> carry <code>Upstream-Status:</code> headers (LLD §A.15)              | Lockfile + patch headers                            |
| D4 | Reproducible build     | Clean checkout on Ubuntu 24.04 / macOS 14: <code>cmake -S . -B build &amp;&amp; cmake --build build</code> produces passing Tier-0 in $< 5$ min                                        | CI pipeline <code>smc-vp-bootstrap.yml</code> green |
| D5 | Firmware sign-off      | Three sign-off firmware images ( <code>smc_boot.elf</code> , <code>smc_flr.elf</code> , <code>smc_telem.elf</code> ) reach <code>tohost = PASS</code> under the VP                     | <code>firmware/signoff/results.json</code>          |
| D6 | Handoff package        | Tagged release <code>smc-vp-1.0.0</code> containing source, PDFs, coverage report, sign-off firmware results, performance numbers, known-issues list                                   | Git tag + release notes                             |

## Gate 5 — Negative Criteria (Explicitly Out of Scope)

The VP exit can be granted with these limitations explicitly recorded; they are *non-goals* for the v1.0.0 release.

| ID | Out-of-scope item                                  | Rationale                                              |
|----|----------------------------------------------------|--------------------------------------------------------|
| N1 | Cycle-accurate fabric arbitration                  | LT abstraction; AT and AT-with-TD are non-goals        |
| N2 | RTL-level CDC, X-propagation, low-power gating     | Domain of the RTL sim, not the VP                      |
| N3 | Physical-layer behaviour of I2C / I3C / AVS PHYs   | Modelled at byte boundaries only                       |
| N4 | Spike interactive debug mode + big-endian builds   | Off by default; require separate gates if ever enabled |
| N5 | I3C DMA mode                                       | <code>MODE_SELECTOR = PIO</code> only (LLD §26)        |
| N6 | SystemC-owned RAM (mode B) for ROM/SRAM as default | Only enabled in dedicated filter/security tests        |

## Phase-to-Gate Mapping

| Phase             | Cluster milestone                         | Gates closed at end of phase         |
|-------------------|-------------------------------------------|--------------------------------------|
| Bring-up          | M1 — scaffold + Spike PLIC                | F1 (P1 only); V1 partial; D1 partial |
| First integration | M2 — SMC TLM PLIC + CLINT                 | F1 complete; V1 complete; V2 partial |
| Full surface      | M3 — SMC Debug Module + P2 IPs            | F2, F3, F4 complete; V2 complete     |
| Sign-off          | M4 — TD fast mode + WFI + commit-log diff | V3, V4, V5; P1–P5; D1–D6             |

# Sign-off Sheet

| Role              | Name | Signature | Date |
|-------------------|------|-----------|------|
| Modeling Lead     |      |           |      |
| Verification Lead |      |           |      |
| Architecture Lead |      |           |      |
| Project Lead      |      |           |      |
| Firmware Lead     |      |           |      |

End of document — `smc-vp-1.0.0` exit criteria.